

FuturLang Chemical Geometry

Clopen Graph Topology of Chemical Knowledge Objects: A Triangle-Rectangle Architecture for Representing, Teaching, and Computing Chemical Compounds

Advanced Topology and FuturPedia Research

A Mathematical Futurology Proposal by **Prophet Wenitte — Solo Founder**

FuturPedia Research Series · FCG-1 · Revised edition

Abstract

Chemistry describes molecules through formulas, diagrams, line notations (SMILES, InChI), coordinates, databases and laboratory records. Each solves a local problem; the fragmentation between them is the real obstacle for education, computation, AI and automation. This paper proposes **FuturLang Chemical Geometry (FCG-1)**, a database-first architecture in which a compound is a structured knowledge object whose core is a typed attributed graph, drawn with a deliberately minimal grammar: **triangles** for entities and **rectangles** for relationships.

The new contribution of this edition is a topological layer. We equip the graph with a finite topology in which *clopen* sets (sets that are simultaneously open and closed) are exactly the unions of connected components. We call a typed graph together with this topology a **clopen graph**. Clopen sets then have a direct chemical reading: they are the *self-contained chemical species* in a system, such as separate molecules in a mixture, ions in a salt, or reactant and product pools. Closed-but-not-open sets model fragments such as functional groups, whose *frontier* consists of attachment bonds. Together with the Betti numbers b_0 (species count) and b_1 (independent ring count), this yields validation rules that are computable, provable and independent of notation.

Status of claims. “Clopen graph” is introduced here as a defined construct, not an established term in the literature. The theorems in §5–§7 are standard facts of finite topology and graph theory applied to this construct; the claims of educational or computational benefit in §12 are hypotheses to be tested empirically.

1. Introduction: Representation Fragmentation

A formula states composition. A structural diagram states connectivity. Coordinates state geometry. A database states relationships and metadata. A manufacturing record states operations. The problem is not missing chemical information but *representation fragmentation*. FCG-1 asks a different first question: not “how do we write this molecule?” but “what is the complete information object for this chemical entity?” The formula then becomes one derived attribute.

Doctrine. Compound $\neq$ Formula, and Formula $\subset$ Compound Knowledge. For aspirin, $C_9H_8O_4$ (molar mass ≈ 180.16 g/mol) is a view of the object, not the object.

2. The Chemical Knowledge Object

A compound is the tuple $C = (I, N, F, S, G, P, R, Q, \Pi)$: identity, names, formula/composition, structure, graph, properties, rules, quality/validation, and provenance. Its mathematical core is

$$M = (V, E, \tau, \beta, P, R, \Pi)$$

where V is the set of entities, E the relationships, $\tau : V \rightarrow T$ assigns entity types (element, functional group, molecule, reaction, batch...), $\beta : E \rightarrow B$ assigns relationship types (single, double, triple, aromatic, ionic, semantic...), P assigns properties to $V \cup E$, R is a set of rules and constraints, and Π stores provenance (source, method, date, version, instrument, confidence).

The triangle-rectangle grammar

Symbol	Meaning	Question answered
$\triangle$ (triangle)	Entity node: atom, functional group, molecule, reaction, experiment, batch	What exists?
$\sqsubset$ (rectangle)	Relationship: bond, containment, constraint, transformation, provenance link	How are entities related?
Nesting	Triangles inside triangles for hierarchy; rectangles inside rectangles for composite relations	At what level of abstraction?

Because every rectangle joins entities, the drawing is a **bipartite incidence graph**: triangles on one side, rectangles on the other, each rectangle adjacent to the entities it relates. This detail is what makes the topology below natural rather than decorative.

3. The Underlying Graph and Its Invariants

Let $G = (V, E)$ with V the entity set and E the relationship set. The adjacency matrix A has $A_{ij} = 1$ when v_i and v_j are connected, or $A_{ij} = b_{ij}$ (bond order) in the order-aware form. Two invariants matter for everything that follows:

$$b_0(G) = \text{number of connected components}, b_1(G) = |E| - |V| + b_0(G)$$

b_0 counts separate species; b_1 (the cycle rank, or “circuit rank”) counts independent rings. For aspirin with explicit hydrogens, $|V| = 21$ atoms ($C_9H_8O_4$) and $|E| = 21$ covalent bonds (13 between heavy atoms, 8 to hydrogen), so $b_0 = 1$ and $b_1 = 21 - 21 + 1 = 1$, which is the single benzene ring. Neither number depends on how the molecule is drawn or serialized.

4. A Finite Topology on a Graph

To speak of clopen sets we need open sets. Treat the graph as a finite poset of “cells”: each edge e lies above its endpoints, written $v \leq e$. Declare a set $U \subseteq V \cup E$ to be **open** when it is closed upward: if $v \in U$ and $v \leq e$, then $e \in U$. This is the Alexandrov (finite, T_0) topology of the face poset, the same construction used for finite models of one-dimensional complexes. The consequences are immediate:

- A set S is **closed** exactly when its complement is open, that is, when $e \in S$ forces both endpoints of e into S . *The closed sets are precisely the subgraphs* (edges never dangle).
- The **closure** of a set adds the endpoints of its edges; the **interior** keeps only those vertices all of whose incident edges are also present.
- The **frontier** (boundary) of a set S is $\text{cl}(S) - \text{int}(S)$. For a subgraph, it consists of its vertices

that still have bonds leaving the subgraph.

Definition (Clopen graph). A clopen graph is a typed attributed graph G together with the face-poset topology $T(G)$ above. A subset $K \subseteq V \cup E$ is **clopen** if it is both open and closed in $T(G)$.

5. The Clopen Characterization Theorem

Theorem 1. A subset K of $V \cup E$ is clopen in $T(G)$ if and only if K is a union of connected components of G (each component taken with all its vertices and edges). Consequently, G has exactly $2^{b_0(G)}$ clopen sets, and G is connected if and only if its only clopen sets are $\{ \}$ and the whole graph.

Proof sketch. Suppose K is clopen. Closed means every edge in K brings its endpoints along; open means every vertex in K brings all its incident edges along. Starting from any element of K and alternating the two rules, one reaches every vertex and edge of its component, so K contains whole components. Conversely, a union of components satisfies both rules. Components partition $V \cup E$ into b_0 blocks, and any subset of blocks gives a distinct clopen set, hence 2^{b_0} . For the final statement, G is connected iff $b_0 = 1$ iff there are 2 clopen sets, $\{ \}$ and G . Q.E.D.

This is the formal core of the proposal: **“is this one molecule or several?” becomes a topological question with a decidable answer.** Clopen sets are the parts of a system that can be separated with nothing cut and nothing left dangling.

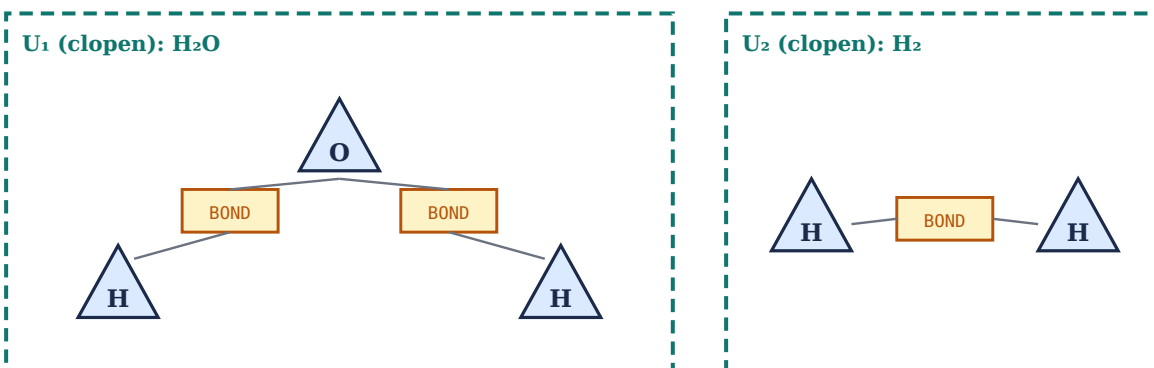


Figure 1. A two-species system in the triangle-rectangle grammar. Dashed regions U_1 and U_2 are the two minimal clopen sets; here $b_0 = 2$, so there are $2^2 = 4$ clopen sets ($\{ \}$, U_1 , U_2 , $U_1 \cup U_2$).

6. Fragments, Functional Groups and the Frontier

Most chemically interesting sets are *not* clopen, and the failure is informative. Consider a functional group F inside a molecule, taken as a subgraph together with its internal bonds. F is closed because its bonds do not dangle. F is generally **not open**, because some atom of F carries a bond to the rest of the molecule. Those bonds are the **frontier edges**: the attachment points of the group.

$$\text{Fr}(F) = \text{cl}(F) - \text{int}(F) = \{ \text{atoms of } F \text{ with at least one bond leaving } F \}$$

Proposition 2. For a closed set F (a subgraph) of a connected graph G , F is clopen if and only if $F = \{ \}$ or $F = G$. Hence every proper, non-empty functional group in a connected molecule has a non-empty frontier. A group with frontier size 1 behaves as a monovalent substituent, size 2 as a linking unit, and so on.

This gives a topological definition of *valence of a fragment* that is independent of drawing style, and it is directly computable: it is the number of edges in G with exactly one endpoint in F . For a carboxyl group in aspirin, the frontier is the single bond to the ring carbon.

Closed, not open: the carboxyl fragment F of acetic acid

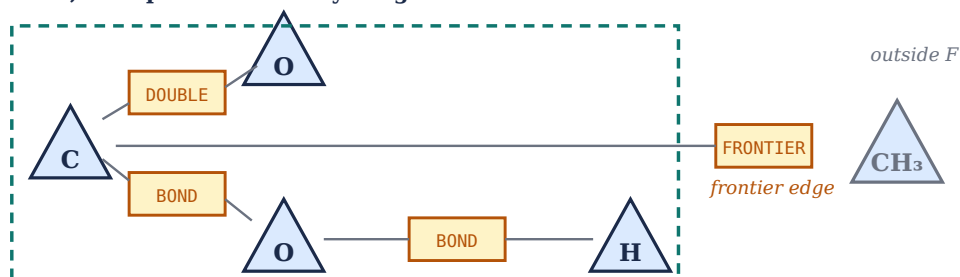


Figure 2. A closed fragment (dashed) is not open: the frontier edge connects it to the rest of the molecule. A clopen set would have no frontier.

7. Chemical Interpretation Table

Topological notion	Chemical reading	Computable test
Clopen set	Self-contained species: molecule, ion, or product pool	Union of connected components
$b_0(G)$	Number of distinct species in the system	Component count
$b_1(G)$	Number of independent rings	$ E - V + b_0$
Closed, not open set	Fragment or functional group with attachment bonds	Frontier non-empty
Frontier of F	Attachment points / open valences	Edges with exactly one endpoint in F
Open, not closed set	Bonds without their atoms: a malformed object	Violates closure; reject at validation
Connected graph	One molecule	Only clopen sets are $\{ \}$ and G

8. Clopen Validation of Compound Objects

The earlier edition defined validity as a conjunction of identity, composition, structure, property and provenance checks. The topological layer sharpens the structural check. A compound object declared as n species must satisfy

$$V_{\text{structure}}(C) \Leftrightarrow b_0(G) = n \wedge \text{every edge has both endpoints in } V \wedge b_1(G) = \text{declared ring count}$$

The middle clause is the closedness of the whole edge set. Composition rules stay as before: the count of vertices with $\tau(v) = C$ must equal 9, with $\tau(v) = H$ equal 8 and with $\tau(v) = O$ equal 4 for aspirin. The distinction between *stored* and *validated* becomes formal: a stored object that fails the clopen partition test is rejected before any downstream use, for instance before conversion to SMILES, where a stray disconnected fragment would otherwise appear silently as a dot-separated component.

9. Isomers: Why Formula Equality Is Not Identity

Two compounds A and B can satisfy $\text{Formula}(A) = \text{Formula}(B)$ with $A \neq B$. Topology makes the separation concrete: isomers have identical vertex-type counts but non-isomorphic typed graphs. The invariants (b_0 , b_1 , degree sequence, frontier sizes of recognized groups) are cheap discriminators; full typed-graph isomorphism (or a canonical form) decides the question. Note that graph isomorphism cannot distinguish stereoisomers on its own, which is why the property map P must carry stereochemical and coordinate information. In logic: $\text{Formula}(A) = \text{Formula}(B)$ does *not* imply $A = B$. A database-first system keeps the distinction by making the graph primary and the formula derived.

10. Higher-Level Entities: Reactions, Batches and Knowledge Graphs

Triangles are not limited to atoms. A triangle may stand for a functional group, molecule, reaction, experiment or batch, and the clopen analysis applies at each level of the hierarchy. A reaction is $R = (S, P, C)$, with reactants, products and conditions; in the incidence graph a reaction rectangle joins reactant species to product species. Within a single system snapshot, reactants and products are clopen sets of the atom-level graph before and after the event. Atom-mapping across the reaction then records *which atoms moved between clopen pools*, so bond breaking and forming are exactly the changes to the component structure of the graph.

Scaling upward, a compound node connects through rectangles to reactions, experiments, instruments, materials and publications, forming a scientific knowledge graph. The principle “no scientific fact without provenance” attaches Π to every property, separating measured, calculated, estimated, imported and proposed values.

11. FuturLang Syntax With Topological Declarations

```
COMPOUND Aspirin {
  ID: "ASPIRIN" NAME: "Acetylsalicylic acid"
  COMPOSITION: C = 9, H = 8, O = 4
  FORMULA: derived from COMPOSITION
  STRUCTURE: NODE C1 ... NODE O4 ; BOND(C1,C2) ...
  GROUPS: CARBOXYLIC_ACID, ESTER // closed, non-clopen fragments
  TOPOLOGY {
    B0 = 1 // one clopen component: a single molecule
    B1 = 1 // one independent ring
    FRONTIER(CARBOXYLIC_ACID) = 1
  }
  PROPERTIES: MOLAR_MASS = 180.16 g/mol [PROVENANCE ...]
  VALIDATION: COUNT(C)=9, COUNT(H)=8, COUNT(O)=4, CLOPEN_PARTITION(n=1)
}
```

The syntax is illustrative, not a finalized language specification. The architectural point is that topological facts are *declared and checked* rather than inferred from a drawing.

12. Education and Computation

Education. The traditional route is Formula $\rightarrow$ Molecule. FCG-1 proposes Object $\rightarrow$ Graph $\rightarrow$ Properties $\rightarrow$ Formula. Students build molecules from triangles and rectangles, then discover that “one molecule vs. a mixture” is a statement about connected components, that rings are counted by b_1 , and that a functional group is a piece with attachment points. This links chemistry to graph theory, topology and database design.

Computation. One semantic object generates many machine representations: formula, SMILES, InChI, adjacency matrix, graph-network input, fingerprint, embedding, JSON, SQL and diagrams. Clopen decomposition is a natural batching step: each clopen component can be processed, embedded or simulated independently, since by Theorem 1 nothing connects it to the rest.

Laboratory and manufacturing. The same grammar represents containers, instruments, batches and tests (for example Batch ---USES-- Material, Batch ---TESTED_BY-- Analysis). A hard rule applies throughout: **representation is not execution**. A compound or reaction object is not an authorization to make or run anything; any automation link remains under validated procedures, qualified equipment, access control, safety and quality systems, and human oversight.

13. Research Hypotheses (Untested)

#	Hypothesis	Suggested test
H1	Students taught with structured molecular objects understand molecular relationships better than students taught formulas alone.	Controlled classroom study with pre/post concept inventory
H2	A common semantic layer lowers integration cost between chemical software systems.	Round-trip conversion study: FCG-1 $\leftrightarrow$ SMILES/InChI/JSON
H3	Explicit component and ring invariants catch malformed records earlier than string-level checks.	Seed errors into a public compound dataset; compare detection rates
H4	Graph-native objects improve interoperability with machine-learning pipelines.	Benchmark feature generation and model inputs from one object
H5	Database-first provenance improves traceability of measurements.	Audit trail completeness on a pilot dataset

These are hypotheses, not findings. None should be treated as established until tested.

14. Roadmap

Phase	Deliverable
I	Formal specification: entity and relationship grammar, type system, ontology, topology module (clopen decomposition, frontier, Betti numbers)
II	Prototype database of compounds, atoms, bonds, groups, properties and provenance
III	Renderer: FuturLang object $\rightarrow$ triangle-rectangle diagram, with clopen regions drawn as dashed boundaries
IV	Import/export: SMILES, InChI, JSON, SQL, XML

Phase	Deliverable
V	AI integration: graph learning, semantic retrieval, classification
VI	Educational platform: build molecules from $\triangle$ and $\sqsupset$, explore components and frontiers
VII	Controlled laboratory-system integration under validated conditions

15. Beyond Chemistry: FuturPedia

The grammar does not depend on chemistry. Gene —EXPRESSES→ Protein, Drug —TARGETS→ Protein, Star —ORBITED_BY→ Planet are all typed graphs, and the clopen analysis again answers “how many independent systems are present?” Within FuturPedia, FCG-1 serves as the chemical module of a wider scientific knowledge language in which documents (PDF, DOCX, HTML, dashboards) are *generated views* of a graph rather than the source of truth, much as compiled outputs are views of source code.

16. Limitations

(1) The face-poset topology is a combinatorial model: it captures connectivity, not bond strength, geometry or electron delocalization, which live in the property map P. (2) Non-covalent association (hydrogen bonds, ionic lattices, coordination) can be modeled as typed relationships, but then the choice of which relationship types count as “cutting” a clopen boundary is a modeling decision that must be declared in the rules R. (3) Hypergraph features, such as multi-center bonds, require extending the incidence graph, which the rectangle already supports as an n-ary relationship node. (4) “Clopen graph” is a proposed term and construct; relation to existing work on finite topological spaces, graph homology and chemical graph theory should be established in formal peer review.

17. Conclusion

FCG-1 treats a compound as a structured knowledge object, drawn with triangles and rectangles, stored as a typed attributed graph, and now analyzed topologically. The clopen graph construct turns three informal chemical ideas into theorems and tests: a *species* is a clopen set, a *fragment* is a closed set with a frontier, and a *ring system* is counted by b_1 . The resulting pipeline is

Object → Graph → Topology → Database → Visualization → Computation → Automation

FCG-1 summary. Primary object: Chemical Knowledge Object. Structure: typed attributed graph with face-poset topology. Grammar: $\triangle$ = Entity, $\sqsupset$ = Relationship. Principle: database object first; visualization is a view of data; one semantic object, many machine representations; understand relationships before memorizing representations.

Prophet Wenitte — Solo Founder

FuturPedia Research · FuturLang Chemical Geometry (FCG-1)